

Science Olympiad RSOI 2022
Dynamic Planet C KEY

___/ 328

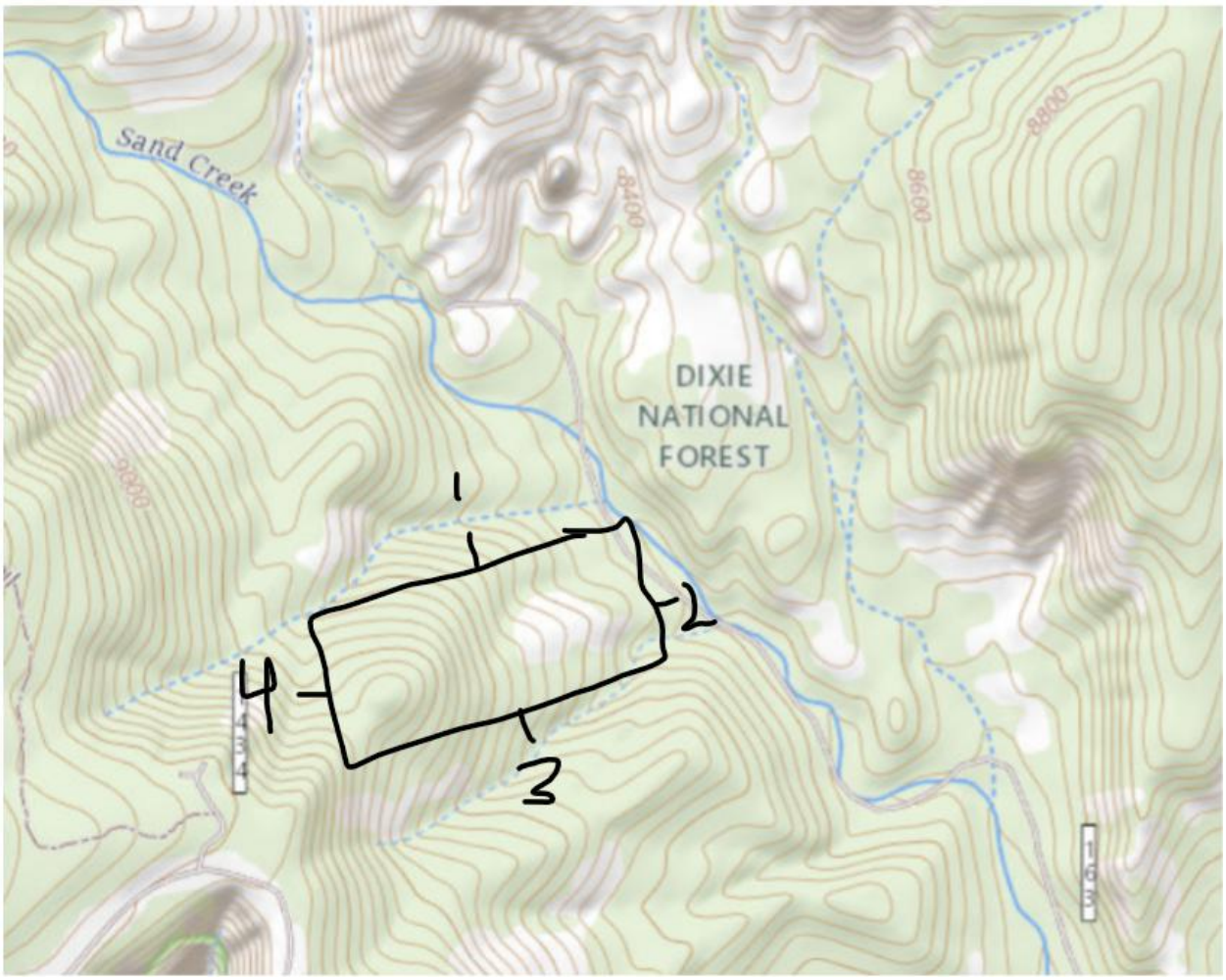
Written by Manank Doshi and Adi Chauhan

Instructions:

- This test has 145 questions, worth a total of 328 points
- Round calculations to 3 significant figures (we won't be too harsh on this, but try to be as accurate as possible)
- You may use an online binder as written in the Event Modifications Document, you may NOT search for answers online
- Showing work is NOT required to gain full credit however we cannot give you partial credit without work.
- Tiebreakers are in the following order: 16, 72, 14, 15, 89, 96, 122
- We hope you enjoy the test and good luck!
- The test is definitely on the longer side and you are not expected to be able to complete the whole test

Section 1: Maps & Graphs & Some Calculations (40 pts)

Maps are cool. In case you get lost while taking this test, we put this section at the start so if you get lost, you can find your way back :)

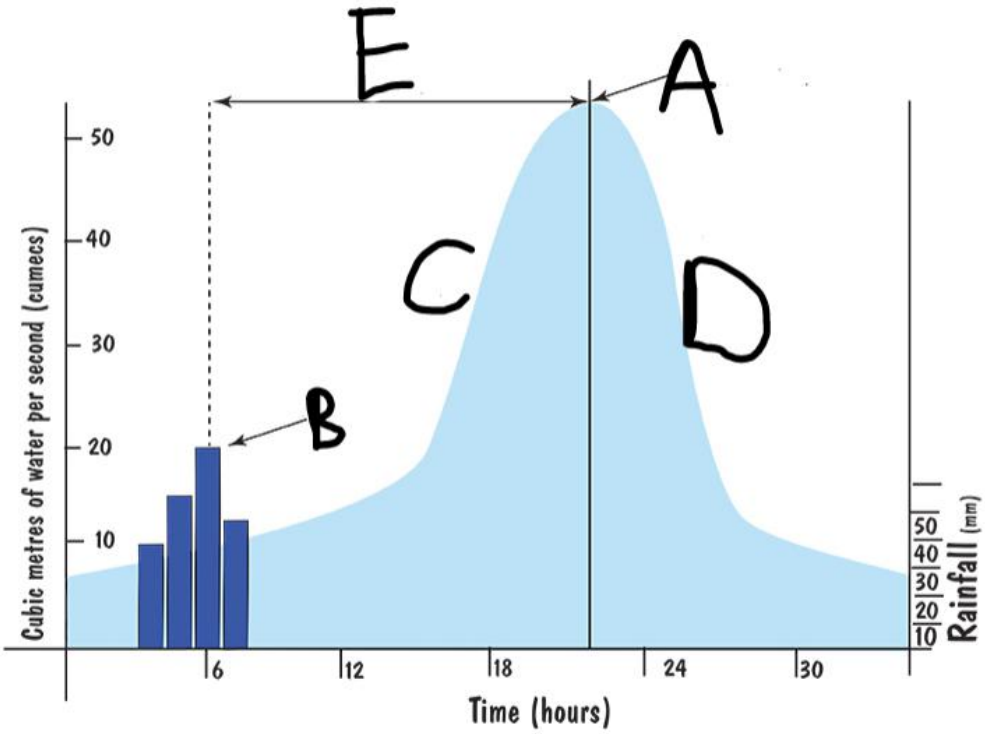


(These are the x,y coordinates in case we need like a better img or something)

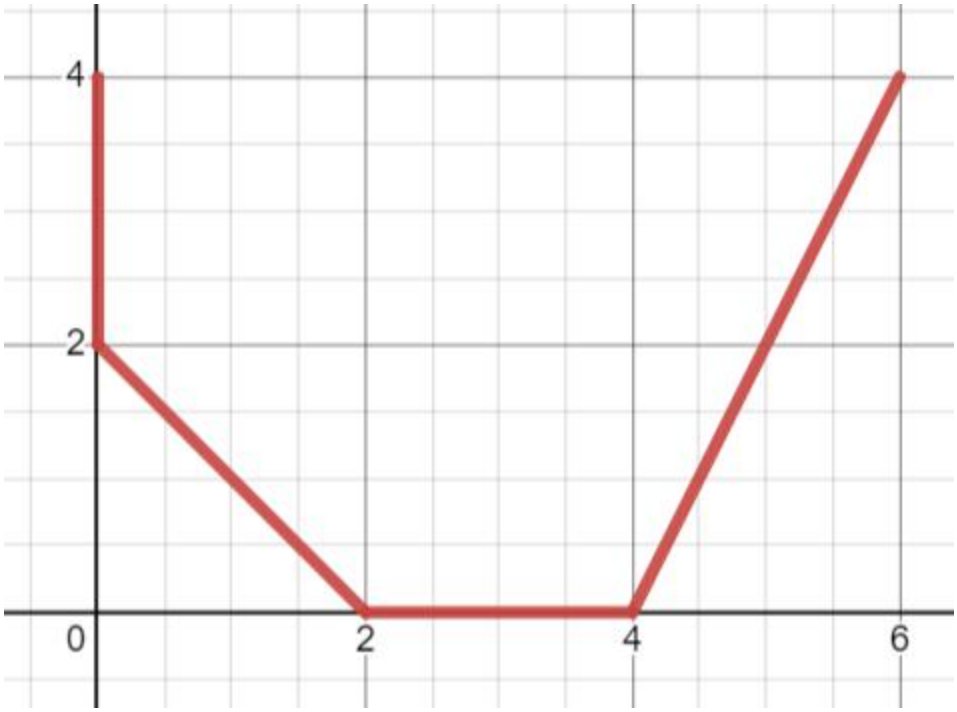
1. [2 pts] What direction (ordinal or cardinal) is the main river flowing, assuming the top is North? How can you tell?
   Southeast, look at the the direction of the v-shapes.
2. [1 pt] What type of creek/stream is Sand Creek?
   Perennial
3. [2 pts] What is the gradient of this map?
   40 ft

4. [1 pt] Based on the contour lines, what can you determine about the region?
   It is a mountainous region with steep slopes
5. [1 pt] Based on the colors of the picture, what can you determine about the region?
   Mostly forest
6. [1 pt] What direction will groundwater flow in the black box?
   a. 1
   b. 2
   c. 3
   d. 4

   This link will be used for the next 2 questions:
   https://drive.google.com/file/d/1m2FATxgshP8pLDiNcSzBr-yZTY--Beef/view?usp=sharing
7. [3 pts] At the very top left, there are some features prevalent in the contour lines, especially around what seems to be like lakes. What are these features? What do these features tell you about the land in that region? You may need to zoom in all the way.
   Depression/Sinkholes, land is most likely made up of limestone, and is most likely karst.
8. [2 pts] At the bottom of the map, there seems to be a wetland. What type of wetland is this in the green areas specifically? Be as specific as possible.
   Wooded marsh or swamp



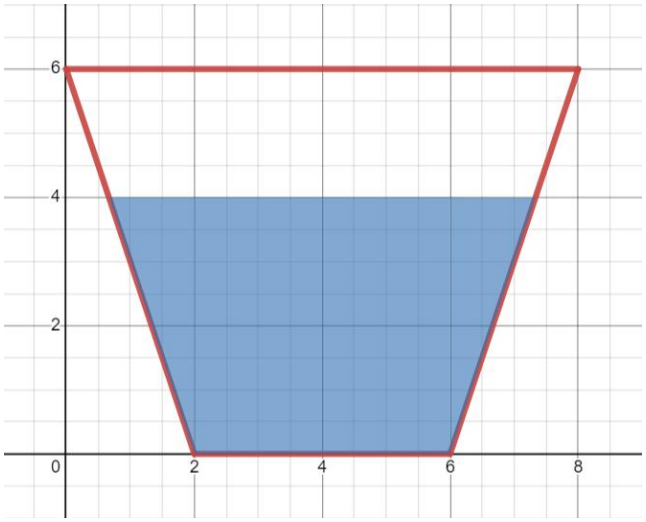
9. [1 pt] What is the name for the diagram above?
   Hydrograph
10. [3 pts] Label parts A-E
    A: Peak flow/discharge
    B: Peak Rainfall
    C: Rising limb
    D: Falling/Recessional Limb
    E: Lag time
    0.5 pts for each, extra 0.5 for getting all correct
11. [3 pts] Pretend that this is a hydrograph of a river near farms in a region which gets abundant rainfall. Would the farmers want a longer or shorter part E? Why? (basically, in the diagram above, would you want a longer or shorter E?)
    Longer, because the shorter the lag time, the faster the river rises, which could cause flooding, so the farmers would want a longer lag time to avoid the flood, aka a flash flood would occur.
12. [2 pts] What 2 graphs/statistics must be combined to create a hydrograph?
    Rainfall/precipitation and discharge/flow



13. [6 pts] The cross-sectional area is shown above in meters. The discharge graph is the hydrograph shown a few questions ago. Find the velocity of the river in m/s at t= 18 hours, t=22 hours, and t=30 hours in m/s.
Cross-Sectional area = 18 m^2. Discharge = velocity * cross-sectional area.
t = 18 hours: Discharge = 40 m^3/s = 18*v, velocity = 2.22 m/s
t=22 hours: Discharge = 52 m^3/s = 18 * v, velocity = 2.89 m/s
t = 30 hours: Discharge = 10m^3/s = 18*v, velocity = 0.56 m/s
14. [5 pts] Let's just do some not-so quick calculations. Stream A has a semi-circle for its channel shape, with a radius of 5m. The straight channel remains full to the top. The channel has a length of 1000m and over the whole length of the channel, the height decreases by 50m. The surface has a roughness coefficient of 1.38. What is the flow rate/discharge of stream A in m/s?

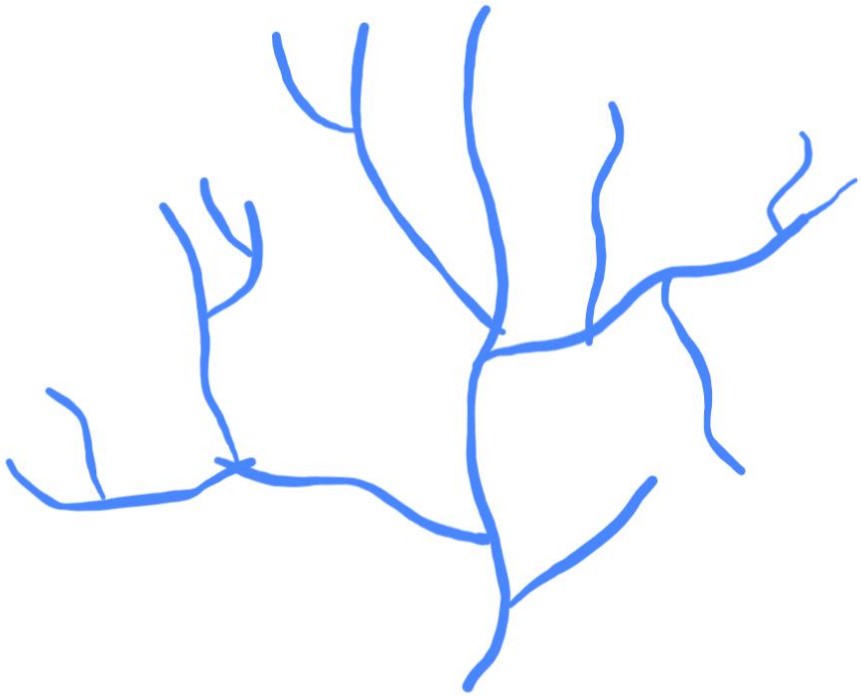
11.72 m^3/s ±0.3; Use Manning Formula: $Q = \frac{K \times A \times \sqrt[3]{R^2} \times \sqrt{S}}{n}$**. K = 1.0, cuz SI units, A = 12.5pi m^2, R = Cross-section/Wetted perimeter = 12.5pi/5pi = , S = 50/1000 = 0.05, n = 1.38. Plug in, you get 11.72 m^3/s.**

15. [7 pts] Stream B has a channel with a shape of a trapezoid and flows parallel to Stream A. What is the discharge/flow rate of Stream B?

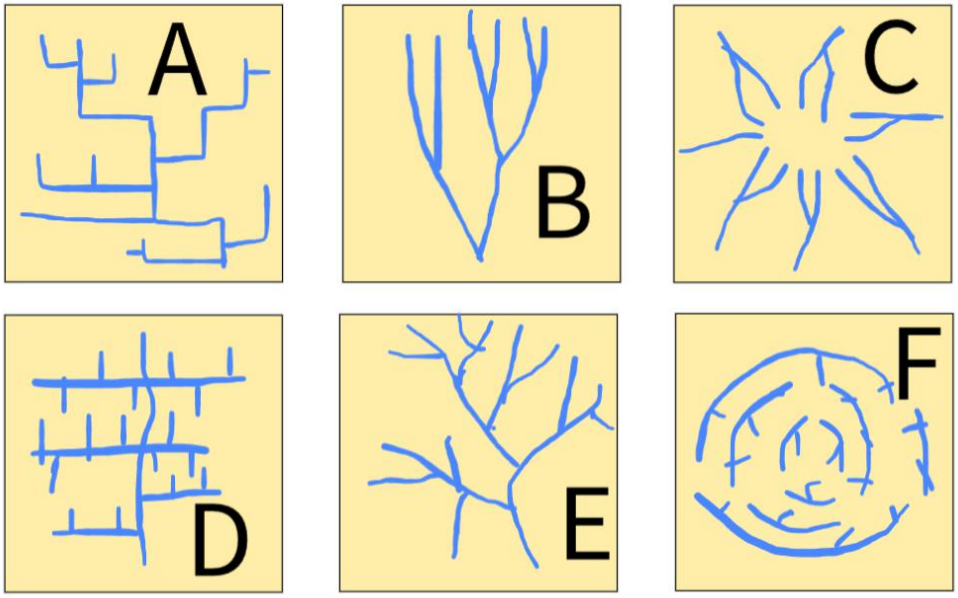


1.11 m^3/s; Use Manning Formula Again. K = 1.0, A= ½(4+20/3)(4) = 64/3, R = (64/3)/(4+4.22+4.22) = 1.715, S = 0.05, N=1.38; Plug in, you get: 1.11 m^3/s

Section 2: Streams & Channels (26 pts)



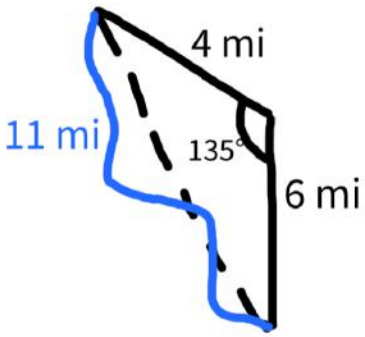
1. [2 pts] What is the Strahler order of the main stream above? (The main stream is the vertical one in the middle flowing from the top of the image to the bottom)
   4
2. [2 pts] What is the Shreve order of the main stream?
   13
   Use the following diagram to answer questions 3-8. Answers must be from A to F, and A-F can be used multiple times or may not be used at all.



3. [1 pt] Which of the above drainage patterns form where bedrock has been eroded in different magnitudes of strength and what is the name of this drainage system?
   D, Trellis
4. [1 pt] Which of the above drainage patterns form when there is a steep slope, such as a mountain range and what is the name of this drainage system?
   B, Parallel
5. [1 pt] Which of the above drainage patterns form when there is a peak like a volcano and what is the name of this drainage system?
   C, Radial
6. [1 pt] Which of the above drainage patterns form where the rock has not been folded, such as on granite rock and what is the name of this drainage system?

E, Dendritic

7. [1 pt] Which of the above drainage patterns form on a region of faulting and basically no topography and what is the name of this drainage system?
   A, Rectangular
8. [1 pt] Which of the above drainage patterns is the most common drainage pattern and what is the name of this drainage system?
   E, Dendritic
9. [1 pt] What do you call the point where a tributary meets the mainstem?
   Confluence
10. [2 pts] What is the major difference between a major drainage divide and a minor drainage divide?
    Major: Water on both sides never meet until ocean, minor: waters meet at a confluence
11. [2 pts] An endorheic basin, also known as a _____ drainage basin, and the water in a basin meets at a single point in a drainage basin, known as a _____.
    Closed, Sink
12. [3 pts] Name 3 catchment factors and describe how they affect the catchment area.
    Topography:
    Shape:
    Size:
    Soil:
    Land Use:
    (0.5 per catchment factor, 0.5 per explanation, accept explanations which make sense)
13. [2 pts] Name 2 specific types of intermittent streams which form in specific climates
    Arroyos, glacial streams
14. [4 pts] Mr. Stream is driving from the beginning to the end of a 11-mile stream. He drives 4 mi directly SE, then drives 6 miles directly South. What is the sinuosity ratio of the river? What type of channel is this?



Use the law of cosines. The straight line distance is 9.27 mi. Sinuosity = 11/9.27 = 1.19. Straight channel

15. [2 pts] How does a braided channel form?
    Sediment bar forms in a channel due to low streamflow or increase in stream load. Water flows onto both sides and that continues to happen, until the channel has many channels and braids.

Section 3: Karst Landscapes (30 pts)

1. [2 pts] What rock makes up Karst landscapes? What is the relation of the ability to form Karst and the purity of this rock.
   Limestone, the purer the limestone, the higher its ability to form karst
2. [2 pts] What are 2 benefits and 1 harm of farming on Karst landscapes?
   Benefits: Fertile soil, plenty of rainfall
   Harms: sinkholes, water escapes quickly through cracks in the rocks.
3. [2 pts] What is a doline karst? Why are the sinkholes in these karst especially problematic?
   Usually rolling plains with few surface streams, and often no valleys on the surface. Sometimes, these sinkholes can lead to heavy property damage because they can be very sudden and abrupt.
4. [1 pt] Name the 2 main harms to Karst formations.
   Contamination and collapse
5. [1 pt] What mineral composes the majority of speleothems?
   CaCO3 minerals, either calcite or aragonite
6. [2 pts] What is the main chemical reaction that occurs in the formation of a Karst landscape?
   H2O + CO2 -> H2CO3

7. [2 pts] Describe the process of the formation of caves.
   Rainwater picks up CO2, and when that rainwater percolates through soil, you get a weak acid, which dissolves the limestone, and over time, some of the joints which were dissolved were large enough to form caves.
8. [4 pts] Let's say, hypothetically, carbon dioxide was removed from the Karst landscape. Why would calcite precipitation be formed if there was excess CaCO3? Give the chemical reaction as well.
   If there is excess CaCO3, the reactants and products have to balance each other out, so to decrease the CaCO3 concentration, it would use this formula: CaCO3 + 2H+ <-> H2CO3 + Ca2+.
9. [2 pts] What is karstification?
   Karstification is the dissolution of soluble rocks by water.
10. [2 pts] Why could pumping groundwater be problematic for karst landscapes?
    Pumping groundwater = lowers the water table. If the water table is lowered to a certain level, then cracks and holes in the landscape are opened up, letting the pressure, which held up the ceiling of the sinkhole, out, creating a sinkhole.
11. [1 pt] What is the major difference between a sinkhole and a karst gulf?
    A Karst gulf is basically a huge sinkhole. It can be hundreds of meters in depth and can contain many smaller sinkholes.
12. [2 pts] What is a Vauclusian spring?
    A spring that originates in typically a cave or shaft system, and the water surges upwards due to high pressure.
13. [2 pts] How do streams "disappear"?
    Streams do not actually disappear, they just flow underground due to things like sinkholes or other Karst features, so we can not see them, which is why it looks like it disappears.
14. [2 pts] How does a losing stream lose water and how does a gaining stream gain water?
    Losing stream loses water because water infiltrates into the ground since the water table is lower than the channel level. A gaining stream gains water because it gains water from the aquifers nearby.
15. [3 pts] If you have 2 regions which may possibly be Karst landscapes and they have the same amount of precipitation, how can you use flora to tell which one has a higher probability of being a Karst landscape,

assuming all else is equal? What does flora indicate about the landscape?

The area with more flora will have a higher probability of being a karst landscape because flora indicate the amount of CO2 trapped, and the more amount of CO2, the more acidic the landscape is, thus a higher probability of being a Karst landscape.

Section 4: River Valleys (40 pts)

1. [2 pts] There are 2 river valleys, let's call them x and y which have the exact same characteristics, except for their age. Mr. Billy Bob wants to do some whitewater rafting through some rapids. If river valley x was formed very recently in the geologic record, while river valley y was formed some time before river valley x, which river valley should Mr. Billy Bob go to for the best places to whitewater raft? Why? Assume that Mr. Billy Bob is immortal and is trying to find the hardest river to whitewater raft out of the 2.

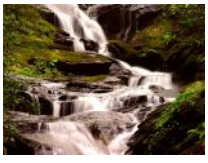
   River Valley X. Younger river valleys tend to have more steep regions and thus more whitewaters and rapids.
2. [2 pts] We once again encounter two different river valleys, this time called A and B. Mr. Billy Bob, tired of rapids, wants a nice, straight path to get to his destination. He knows that river valley A has uniformly resistant bands, while river valley B does not have uniformly resistant rock bands. Which one of these should Mr. Billy Bob take to get the straightest path? Why?

   River Valley A, uniformly resistant bands means that the river will flow smoother and curve less, so will be straighter.
3. [3 pts] So, it actually turns out River Valley B and River Valley Y (now let us call it River Valley BY) were the same river valley. How would you characterize the river's gradient at the source and at the base level based on the characteristics of River Valley Y and River Valley B? The river's highest gradient is 100ft/mi. Express your answer in limits where source is -infinity and base level is infinity, x is the position of the river in miles, and f(x) is the gradient in ft/mi. You can just write a complete sentence using limits, no need for limits notation. (CaLCulUs! Or Precalc! lol)

   At the source, there will be a steep gradient. At the base level, there will be a low gradient approaching 0. In limits: as x approaches

-infinity, f(x) approaches 100. As x approaches +infinity, f(x) approaches 0.

4. [2 pts] Hypothetically, river Valley BY actually is the river valley of the Mississippi River. What is the base level of this river? The answer is NOT a number.
   Sea level
5. [3 pts] During the path of river valley BY, it encounters a region with little vegetation, and right after, there is a region with a lot of vegetation. Both regions are exactly the same besides the vegetation content. How would the bank erosion change from the first region to the second region or would it even change? Why or why not?
   Bank erosion in the first region would be higher than the second, because roots of the vegetation increase strength of the soil or whatever material the bank is made up of, so it will be less likely to erode.
6. [2 pts] How are terraces formed?
   a. **Downcutting/carving down**
   b. Sea level rise
   c. Constant flooding
   d. Upcutting
7. [6 pts] Identify the type of waterfall

 1	 2	 3	 4	 5
 6	 7	 8	 9	 10

1. **Fan**
2. **Moulin**
3. **Segmented**
4. **Punchbowl**

5. **Ledge/Block/Cataract**
6. **Tiered**
7. **Horsetail**
8. **Cascade**
9. **Plunge**
10. **Slide**

0.5 pts per correct, +1 for all correct.

8. [3 pts] A particle of alluvium travels all the way from rest to an alluvial fan. Describe the process for how this occurs and what regions/places the particle has to pass through.
   Gets picked up by a flowing body of water (a river or stream) and is transported to the alluvial fan. The particle has to travel through the feeder canyon, past the apex (or narrow point), and then is spread out on the fan.
9. [1 pt] What is a floodplain?
   a. **Area of land near a stream/river**
   b. Plains/Priaries when, with heavy rain, can lead to flash floods
   c. Area of land surrounding a lake
   d. Beaches (near the ocean)
10. [2 pts] Reelfoot Lake, found in Tennessee, is a shallow oxbow lake created by an earthquake. How can an earthquake form an oxbow lake?
    Earthquakes cause land changes, such as causing the river to change path.
11. [3 pts] There are 3 types of bedding which can be found in a simple delta: botomset, frontset, topset. Order these 3 types of bedding in order of finest to coarsest.
    Bottomset, Topset, Frontset
12. [3 pts] Oxbow lakes are formed when the bends are cutoff for a shorter path, but how do these cutoffs/chutes occur?
    Lateral erosion of the bank causes the stream to cut through the neck of the loop. Streams flow through chutes with low gradients only during high-water periods, but high-gradient chutes divert the stream, leaving the former meander loop as an isolated oxbow lake.
13. [4 pts] How do levees form naturally? Nam 1 advantage of creating levees artificially, and 1 disadvantage of creating levees artificially.
    Natural Levees form when sediment is pushed to the side of the river and piles up. Advantage: can make levee as strong as possible

(accept any others which make sense). Disadvantage: Can be very expensive (accept any others which make sense)

14. [2 pts] Point bars generally reflect what characteristic of a river? What happens to the point bar when the river is flooded?
    The capacity. When flooded, the point bar can accumulate debris and driftwood and can also be flooded.
15. [2 pts] Let's finish this section with an easy question. How do the meanders in a meandering stream form?
    Meanders are produced when water in the stream channel erodes the sediments of an outer bend of a streambank and deposits this and other sediment on subsequent inner bends downstream.

Section 5: Lakes (50 pts)

1. [1 pt] The world's largest artificial lake by volume:
    a. Lake Volta
    b. Manicouagan Réservoir
    c. Lake Kariba
    d. Lake Nasser

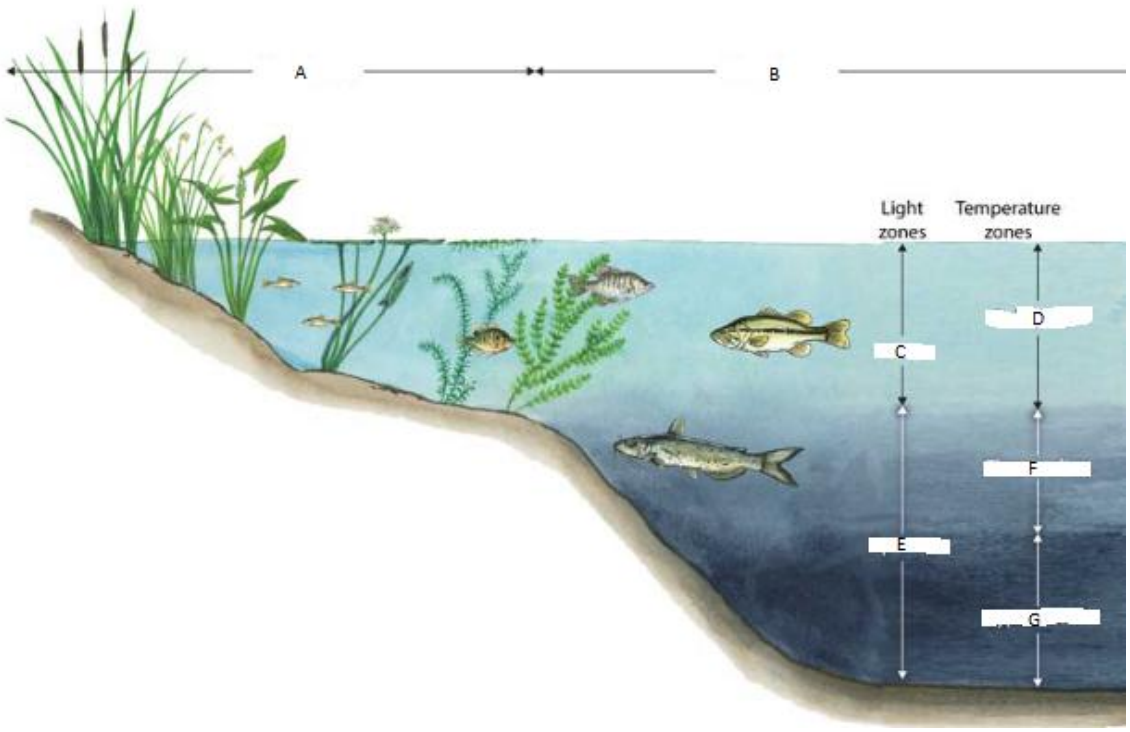
2. [2 pts] Describe the formation of a tectonic lake.

Formed by the deformation and resulting lateral and vertical movements of the Earth's crust. These movements include faulting, tilting, folding, and warping

3. [1 pt] What percent of Earth's liquid surface freshwater is contained in lakes?

87% (+- 5%)

4. [7 pts] Label the parts of the lake zone diagram



A. Littoral Zone
B. Limnetic Zone
C. Photic
D. Epilimnion
E. Aphotic
F. Thermocline
G. Hypolimnion

5. [6 pts] Compare and contrast a cold monomictic lake with a warm monomictic lake. Give as many details as possible

Cold monomictic lakes are lakes that are covered by ice throughout much of the year/During their brief "summer", the surface waters remain at or below 4 °C/The ice prevents these lakes from mixing in winter/During summer, these lakes lack significant thermal stratification, and they mix thoroughly from top to bottom/These lakes are typical of cold-climate regions

Warm monomictic lakes are lakes that never freeze, and are thermally stratified throughout much of the year/The density difference between the warm surface waters (the epilimnion) and the colder bottom waters (the hypolimnion) prevents these lakes from mixing in summer/During

winter, the surface waters cool to a temperature equal to the bottom waters/Lacking significant thermal stratification, these lakes mix thoroughly each winter from top to bottom/These lakes are typical of warm tropical regions.

6. [1 pt] What type of lake is the image below?



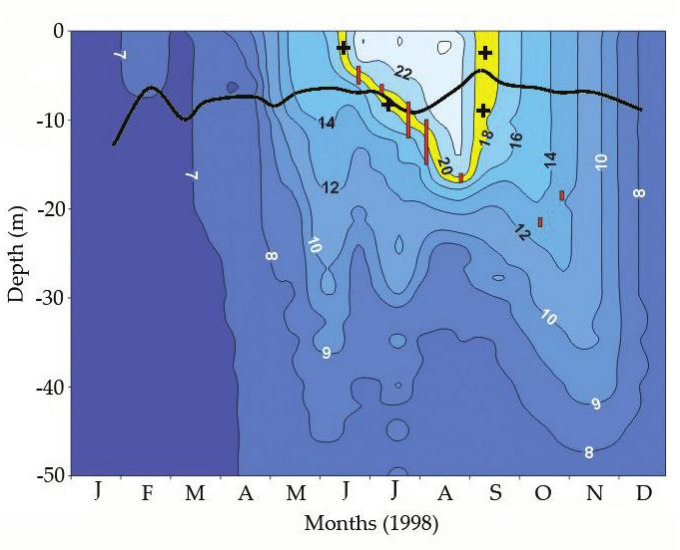
Solution lake

7. [2 pt] What is the type of lake below and how does it form?



Oasis lake, formed through an underground aquifer/river. Pressure forces the water to go up and with enough water, there is an oasis.

8. [4 pts] What is the relationship between calderas and the formation of lakes? What is the name of this type of lake? Give an example of it.
When a volcano erupts, it often forms a caldera. Over time, water gathers in these calderas, creating a volcanic crater lake. One example is Crater lake, check google for more.
9. [4 pts] There are 3 lakes. Lake A has one inflow and one outflow stream. Lake B has one outflow stream. Lake C has one inflow stream. All lakes have the same characteristics besides the inflow and outflow. Assume that the inflow of all lakes are equal to the outflow of all lakes. How will the level of water change in each of the lakes in relation to each other? What is the primary source of the loss of water in each?
Lake A will have no change in the level of water, lake B will lose water so will have a decrease in the surface level, and lake C will gain water so will have an increase in the surface level. The primary source of loss of water for Lakes A and B is the outlet, while for C it is evaporation and infiltration
10. [3 pts] Why can thermal stratification lead to problems for aquatic plants in the hypolimnion in deep lakes?
The hypolimnion can often be depleted of oxygen since it is located at the very bottom of the lake. Since the hypolimnion has very little contact and transfer with the surface, it is hard to gain oxygen. The aquatic plants need oxygen, and if they do not get it, they can suffer.
11. [5 pts] Rank the following 4 trophic classifications in order of increasing productivity: Hypertrophic, Mesotrophic, Oligotrophic, Eutrophic. Explain how you know using Secchi depth and what Secchi depth measures.
Oligotrophic, Mesotrophic, Eutrophic, Hypertrophic. Secchi depth is used to measure water transparency. When the Secchi depth is greatest, which is at Oligotrophic, you can see the farthest because there are no algal blooms or much other productivity blocking the sight.



12. [1 pt] This diagram is an isotherm of Lake Maggiore. The lines are in degrees celsius. How would you classify this lake in terms of stratification?

A. Monomictic
B. Amictic
C. Dimictic
D. Polymictic

13. [6 pts] Give a quick explanation of how each process can lead to lake formation. The processes are glaciation, rifting, and faulting.

Glaciation: When glaciers move forward, they erode. When they recede, they leave depressions and holes, and the meltwater fills up these areas
Rifting: formed as a result of subsidence related to movement on faults within a rift zone
Faulting: A depression forms when a weak section of the earth's crust separates, resulting in an earthquake. Rainfall and groundwater may collect in this depression, forming a lake.

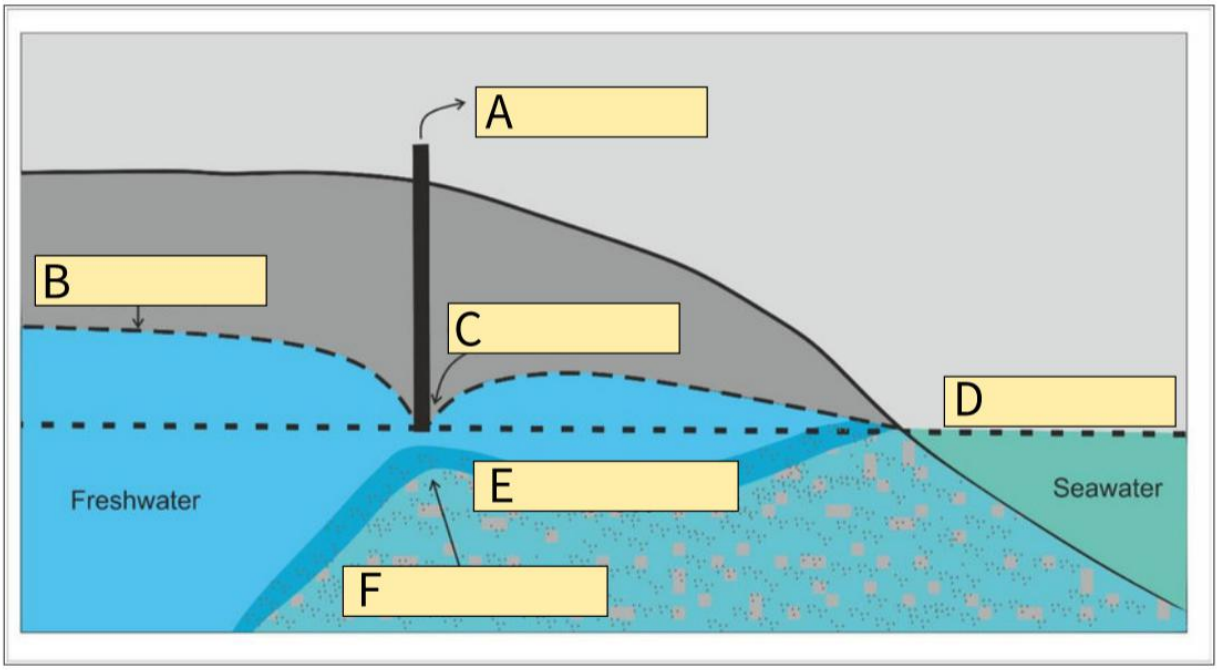
14. [3 pts] What is a seiche and how is it formed?

A seiche is a wave in a closed body of water, often found on lakes. Seiches form when there are strong winds and rapid changes in atmospheric pressure.

15. [4 pts] What are the 2 main reasons why freshwater acidifies? Explain how these reasons acidify water.

Atmospheric Depositions: the process, long recognized by scientists, whereby precipitation (rain, snow, fog), particles, aerosols, and gases move from the atmosphere to the earth's surface.
Soil Leaching: Excess water removes water-soluble nutrients out of the soil, by runoff or drainage. Leaching is an environmental problem for agricultural professionals whether chemical-heavy fertilisers or chemicals are swept out and find their way into water bodies.

Section 6: Groundwater (38 pts)



1. [6 pts] Label parts A-F in the diagram above.
   A: Well
   B: Water Table
   C: Cone of Depression
   D: Sea Level
   E: Freshwater-Seawater interface (or anything similar to this)
   F: Seawater/saltwater Intrusion
   1 pt per part
2. [3 pts] This isn't your normal groundwater/aquifer diagram, but rather a diagram of a specific type of groundwater interaction. What is the specific type of aquifer shown here? How can you tell?
   Coastal Aquifer, freshwater and seawater mixing

3. [2 pts] What effect does excess phosphorus have on the quality of groundwater? Explain.
   Hurts the quality, too much phosphorus means too many nutrients for plants and animals, meaning too many plants and algae, leading to low dissolved oxygen concentrations and hurting the animals. Basically, eutrophication.
4. [3 pts] Why is laminar flow characterized as parallel lines? Name 2 characteristics of laminar flow as well.
   Laminar flow tends to be at slow velocities, so the water does not mix as much. There are very few currents, so little mixing. Also, there is high momentum diffusion and reduced movement convection, which is caused by laminar flow.
5. [2 pts] What is the recharge rate in m of a freshwater lens with precipitation of 10m and evapotranspiration of 8m?
   2 m, 10m-8m = 2m
6. [2 pts] A certain region uses a well, but overuses this well. What will the community need to do to this well to continue using this well? Why?
   The well will need to be dug deeper because using too much of the groundwater will cause the water table to decrease, so to maintain the same amount of water being used (although they should save water), they will have to make the well deeper and deeper.
7. [2 pts] How does water exactly move up in an artesian well if gravity pulls the water down?
   Continuous penetration of water and pressure from the water's weight, aka Hydrostatic pressure pushes the water up to the surface of the well.
8. [2 pts] On which of the following can a perched water table be formed? Select all that apply
   a. **Aquiclude**
   b. **Aquitard**
   c. Aquifer
   d. Artesian well
   e. Piezometric Surface
9. [1 pt] If a perched water table intersects a valley wall, how/from where will the water be discharged?
   Spring

10. [3 pts] Aquifer R is an unconfined aquifer and Aquifer P is a confined aquifer. What is the difference between a cone of depression in Aquifer R and Aquifer P?
    Aquifer R has an actual depression in the level of water, while Aquifer P just has a reduction in the pressure head.
11. [1 pt] Over time, aquifer R has 2 cones of depressions which intersect each other. How does this affect the water level of the aquifer, if at all?
    a. No change
    b. Increase
    c. Decrease
12. [3 pts] In terms of flow patterns, how is the use of Darcy's formula limited? If we have a short period of time, what do we add to Darcy's formula to account for this?
    Darcy's law can only be used in laminar/parallel patterns, not turbulent. In short periods of time, we need to add a time derivative.
13. [4 pts] What is the specific discharge in m/s of an aquifer with a hydraulic gradient of 0.25 and passes through medium sand, which has a hydraulic conductivity of 3 m/d?
    8.68 x 10^-6 m/s; Darcy's Law: q=Q/A = -K*dh/dl. q is the specific discharge, hydraulic gradient is dh/dl which is 0.25, and hydraulic conductivity is 3 m/d = 3.472 x 10^-5 m/s. 0.25 * 3.472 = 8.68 x 10^-6 m/s.
14. [1 pt] Which of the following defines the zone of saturation or is another name? Choose all that apply.
    a. Phreatic Zone
    b. Vadose Zone
    c. Pores in rock/soil contain air and water
    d. Pores contain only water
15. [3 pts] There are 2 aquifers, aquifer C and aquifer D. They both are located in the same region and the water company is deciding which groundwater to get water from. If the main difference is that aquifer C is much older than aquifer D, from which aquifer should the water company get the water from to have the chance of getting the cleanest/healthiest water? Why?
    Aquifer D. Over time, more and more material gets dissolved into the aquifer. If there are dangerous elements which can contaminate the groundwater, aquifer C would have more of these elements, so the water would be the most contaminated.

Section 7: Sediments & Stream flow (30 pts)

1. [2 pt] Rank the stream load from largest to smallest: bed load, dissolved load, suspended load.
   Bed load, suspended load, dissolved load.
2. [3 pts] A sandstone rock has a volume of 27 cubic centimeters and voidspace of 6.4 cubic centimeters, what is the porosity of the sample? Round to the nearest hundredth place.
   0.24 or 24%
3. [3 pts] Consider the following rivers:
   River 1: Width at mouth is 20m, depth at mouth is 20m, max discharge is 100 m^3/s.
   River 2: Width at mouth is 300m, depth at mouth is 10m, max discharge is 1000 m^3/s.
   River 3: Width at mouth is 50,000m, depth at mouth is 20m, max discharge is 100,000 m^3/s.
   Which has the highest stream competence?
   River 1
4. [2 pts] Explain the difference between permeability and porosity
   Porosity is how much space there is in a rock to hold water.
   Permeability is a measure of how well water can flow through rock.
5. [5 pts] There is a semicircle cross section of a river. The max depth is 5.6 m, the water velocity is 7.9 meters/second. What is the stream flow of this location in m^3/sec? (units not required in answer box)
   389.15 (± 1 m^3/sec)



6. [1 pt] What type of channel is in the above image?

a. Braided
b. Straight
c. Meandering
d. Hybrid

7. [1 pt] What is the estimated sinuosity of the channel in the above image?
a) 2.0
b) 2.7
c) 3.5
d) 7.4
8. [3 pts] Two cities are 30.7 miles apart with a stream connecting them. If a raft floats down the stream between the towns, it will have traveled 34.9 miles. What is the stream's sinuosity?
1.1
9. [1 pt] What type of channel is in question 8?
Straight
10. [2 pt] What is a braided channel and how are they formed?
Channel divided into smaller channels, formed by temporary islands (eyots) as a result of increased sediment load.
11. [1 pt] A clast of diameter 20 mm is classified as
a. Sand
b. Pebble
c. Boulder
d. Silt
12. [2 pts] What are the typical sediment types in an aeolian environment?
a. Gravel
b. Sand
c. Clay
d. Silt
e. Silica mud
f. Carbonates
13. [2 pts] What is the difference between clastic and chemical sediments?
Clastic = solid fragments
Chemical = material transported as ions in a solution
14. [1 pt] What are the largest particles a stream can transport if its velocity is 12 cm/sec?
Silt
15. [1 pt] Stream erosion occurs the most in ___

a. The outer bend of the stream
b. The end of the stream
c. The inner bend of the stream
d. The source of the stream

Section 8: Wetlands & Critical Zone Hydrology (28 points)

1. [2 pts] Which of the following forces cause infiltration?
   a. Matric forces
   b. Capillary
   c. Friction
   d. Gravity
   e. Air resistance
2. [2 pts] What is a similarity between hydraulic and hydrostatic pingos?
   Both result from advancing permafrost/Both form an ice lens
3. [1 pt] Mangrove forests are an example of what type of wetland?

a. Marsh
b. Fen
c. Bog
d. Swamp

The following image is used for questions 4-6



4. [1 pt] Identify the wetland above

Bog

5. [1 pt] Explain the formation of the wetland

Accumulates peat as a deposit of dead plant materials often mosses. Occur where the water at the ground surface is acidic and low in nutrients

6. [1 pt] What is the main water source?

Water derived mainly from precipitation.

7. [2 pts] Explain the difference between a marsh and a swamp.

Marshes contain herbaceous rather than wooded plant species, while swamps are forested. Marshes have mainly grasses and reeds, swamps have trees and shrubs.

8. [1 pt] What are two factors differentiating wetlands from other terrestrial landforms?

Characteristic vegetation of aquatic plants, adapted to the unique anoxic hydric soils/constantly flooded, allowing anoxic conditions to prevail in soils.

9. [2 pts] Wetlands can be classified as 3 things based on the hydrology: a) associated with streams, b) associated with lakes and reservoirs, and c) isolated. What is the scientific name for each?

a. **Riverine**
b. **Lacustrine**
c. **Palustrine**

10. [2 pts] What is a vernal pool? Why are they often overlooked when talking about wetlands?

A vernal pool is a seasonal pool of water, often providing habitats for plants and animals. They are often overlooked because they are so small and only are flooded for short periods of time.

11. [3 pts] Areas A and B are two areas right next to each other with similar characteristics except that area B has a smaller region made up of wetlands than area A. During a flood, which area would you rather leave in to face the least effects? Why?

Area A because more wetlands means more water can be absorbed, so less effects of flooding.

12. [3 pts] What is permafrost? If all the permafrost melted/thawed from Earth, how would that affect the Earth's climate?

Permafrost is ground that remains frozen for usually 2 or more years. If all permafrost melted from the Earth, greenhouse gases would be released, leading to more global warming.

13. [2 pts] How does a saltwater marsh form?

Salt marshes generally form in coastal areas that are relatively sheltered from harsh ocean waves and where rivers or creeks deposit a special type

of fine sediment. These areas of fine sediment are referred to as mud flats. As the sediment continues to collect, these flats grow in size and elevation. Flooding of the area becomes less intense due to the increase in elevation, which allows plants to move into the area. These plants reduce the speed at which the river or the creek flows into the ocean, allowing more sediment to settle. As the amounts of river sediment and plant species increases, the amount of sediment retained from high tide also increases. Over time, other plant species colonize the area.

14. [2 pts] What is the boundary between the upper wet part of the soil and the dry lower part?

Wetting front

15. [3 pts] What region has the greatest concentration of pingos?

Tuktoyaktuk

Section 9: Land changes, Hydro cycle, and Pollution (36 points)

1. [4 pts] Classify the following as point or nonpoint source pollution. Enter P for point source or NP for nonpoint source.
    a. Sewage plant leaks chemicals
    b. Runoff from streets
    c. Animal waste from ranches
    d. Chemical rain
    e. Ship has an oil leak
    f. Forest
    g. Pesticides
    h. Tire factory

    A: P
    B: NP
    C: NP
    D: NP
    E: P
    F: NP
    G: NP
    H: P
2. [2 pts] Besides evaporation, what other process can lead to more moisture in the atmosphere? Why is this process less common?

Sublimation, less common because sublimation goes from solid to gas, and usually passes through liquid.

3. [3 pts] What is the process called when water falls from clouds in a liquid form? (If you get this wrong, we will be very sad :( ) Why does liquid water fall to the ground from clouds?
   Precipitation happens when water droplets grow larger and larger, they get too heavy for clouds and fall to the ground.
4. [1 pt] What is evapotranspiration?
   Sum of evaporation and transpiration.
5. [2 pts] Order the following processes from how water starts in the ocean and ends up back in the ocean: Runoff, Condensation, Evaporation, Precipitation.
   Evaporation, Condensation, Precipitation, Runoff
6. [2 pts] Once water hits the ground, which of the following can happen to the liquid water before it goes into the ocean? Choose all that apply.
   a. **Runoff**
   b. **Evaporate**
   c. **Infiltrate**
   d. **Plant uptake**
   e. Desublimation
7. [2 pts] How could nuclear power plants lead to the pollution of water? (The power plant does not blow up or anything like that lol)
   Nuclear power plants rely heavily on water for cooling, and during the cooling process, water comes in contact with radioactive substances. If not disposed of correctly or not filtered properly, the water could threaten groundwater, surface water, and other water sources.
8. [3 pts] What is it meant by “anthropogenic” contamination? Which of the following in question 1 of this section are anthropogenic?
   Anthropogenic = human made/created. A, B, C, E, G, H
9. [2 pts] What is the leading type of contaminants of freshwater surface lakes and rivers in the US?
   Nutrients (nitrates, phosphates, etc.)
10. [3 pts] Flooding can be very destructive and change the landscape completely. However, flooding can be beneficial occasionally to certain regions, although it is destructive. How could flooding be beneficial to the groundwater of a certain region? How could flooding be detrimental to the groundwater of a region? Name 1 for each.

Beneficial: Recharge groundwater, bring nutrients to the soil.
Harms: Bring contaminants to the groundwater
If any other reasonable answer, give credit

11. [2 pts] A certain company would like to divert a stream, but knows that there could be consequences to the stream. Name 2 of these consequences.
    Reduced water levels, hurt water quality, change characteristics of the channel/stream, accept any other reasonable answer.
12. [2 pts] What is sedimentation in hydrology?
    Particles/sediment sinks to the ground due to gravity
13. [3 pts] Order the following precipitation sizes from smallest to largest: cloudburst, excessive rain, drizzle, mist, moderate rain, heavy rain, fog, light rain.
    Fog, Mist, Drizzle, Light rain, Moderate rain, Heavy rain, excessive rain, cloudburst.
14. [2 pts] Why does runoff occur if water should simply be infiltrating the ground?
    The ground is already saturated, so more water cannot infiltrate, so the water has no choice but to runoff.
15. [3 pts] In a period of 50 years, there were 6 floods in the town of ScIEncE OLyMpiAD. What is the recurrence interval for the 3rd worst flood?
    Formula: 17%. R=(years+1)/rank. R = (50+1)/3 = 17%

Section 10: Trivia! (10 pts)

This section is basic trivia about hydrology, worth a total of 10 points.

1. [1 point] What is the largest freshwater lake in the world by surface area? How about by volume? (We're gonna say that the Caspian Sea is not a lake)
   Lake Superior, Lake Baikal
2. [1 point] What percent of Earth's surface is covered by water? What percent of water is freshwater? So, what percent of Earth's surface is covered by freshwater (Round to nearest whole number for first part, tenth for the last 2 parts, and do not include percent sign)
   71%, 2.5%, 1.8%
3. [1 point] What process occurs when water in clouds becomes rain?
   Condensation

4. [1 point] What is the largest watershed in the US? Through how many US states does this watershed pass through?
   Mississippi River Watershed, 31 states
5. [1 point] What percent of the Earth's land surface is made up of Karst?
   Between 8-15%
6. [1 point] Which country has the most freshwater lakes in the world?
   Canada
7. [1 point] What are the 3 longest rivers in the world by length? Rank them in order of longest to shortest.
   Amazon, Nile, Yangtze
8. [1 point] Which US state has the most rivers?
   Alaska
9. [1 point] What is the smallest lake of the Great Lake by volume? By surface area? Answer in that order.
   Lake Erie, Lake Ontario
10. [1 point] Which US state had the least precipitation in 2021 and is the driest state?
    Nevada